

Analyzing the Network of Disease Distribution by Country and Distance

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Abstract

Infectious disease surveillance accumulates a large relational dataset: every reported outbreak links a country to a disease and a year. This paper constructs a global network from outbreak records spanning 233 countries and seventy diseases between 1996 and 2022 and analyzes it along two complementary axes. The temporal axis tracks how centrality, the influence and connectivity of individual countries, evolves year by year, revealing a network that is structurally resilient (a maximum k-core of 221) but whose connectivity surges sharply during major pandemics. The geospatial axis weights edges by geographic proximity and tests directly whether distance shapes the resulting structure. A Mantel permutation test, the appropriate tool for correlating two matrices defined over the same set of nodes, shows that geographic distance is significantly associated with disease co-reporting (Pearson $r = -0.192$, permutation $p < 0.0001$ over 9,999 permutations), a relationship that holds, and is in fact stronger, when the COVID-19 pandemic years are included. Because geographically close countries are also often well connected by direct flights, a partial Mantel test controlling for air-travel connectivity confirms the distance effect survives, weakening only slightly (partial $r = -0.176$, $p < 0.001$). Stratifying this confound-controlled test by disease transmission mode shows the distance effect is weakest for vector-borne disease and strongest for respiratory disease, a pattern consistent with environmental filtering by global climate zone overriding simple geographic proximity for vector-borne disease distribution. These results give the spatial intuition behind global disease surveillance a quantitative foundation and identify Germany, Russia, and the small island and territorial states of the Caribbean as structurally distinct types of nodes, central hubs in one case, geographically isolated outliers in the other, that warrant different surveillance strategies.

Keywords: network science, infectious disease, centrality, k-core decomposition, Mantel test, partial Mantel test, geospatial analysis

1 Introduction

Infectious diseases continue to pose a global health challenge, causing widespread morbidity and mortality despite advances in medical technology, vaccine development, and public health awareness. Increasing global mobility and the resulting complexity of interaction between populations make the spread of infectious disease harder to manage and harder to anticipate [1]. This paper applies network science to the spread of infectious disease across countries, explicitly incorporating geographic distance into the analysis. Network science has already demonstrated its value in biology, neuroscience, and health science more broadly [2, 3, 4], and the same tools transfer naturally to disease surveillance, where the relevant question is not just which countries report a disease but how those countries are connected to one another through shared outbreaks.

1.1 Significance of the Study

Infectious diseases remain among the ten leading causes of death worldwide [5]. Their persistent threat, combined with their capacity to spread rapidly across borders, motivates analytical approaches that characterize disease dynamics at a scale individual case reports cannot. Network science converts a large table of country-disease-year records into a structure whose shape, which countries are central, which are peripheral, which regions cluster together, can be read directly off the data.

This matters for two practical reasons beyond the descriptive value of the network itself. International disease surveillance operates under fixed resources, and a tool that identifies which countries function as structural bridges, rather than simply which countries report the most cases, gives surveillance planners a way to prioritize monitoring and response capacity toward locations whose loss would disconnect otherwise distant parts of the network, not just locations with high case counts. Betweenness centrality and k-core membership carry exactly this kind of structural signal, and neither is visible from country-level case counts alone. The role geography plays in this structure is also shifting: global air travel and mobility already reshape how diseases move between regions relative to simple geographic distance [6], and climate change is expanding the geographic range of vector habitats into regions that were not previously climate-suitable [7]. Any distance-based account of disease structure measured today is therefore a snapshot of a relationship likely to shift over time, and establishing statistically how strongly geography currently predicts disease co-reporting, and whether that relationship survives controlling for travel connectivity, provides a quantitative baseline against which future shifts driven by mobility and climate change can be measured.

The network built here connects countries rather than individuals. An edge records that two countries reported the same disease in the same year, a different claim from a confirmed transmission event linking two specific cases. Person-to-person contact networks and mobility-based transmission models operate at a finer resolution and answer a different question: how a disease moves from one infected person to the next [6]. The country-level network instead serves disease surveillance at the resolution surveillance data actually has available, standardized and openly reported across 233 countries and seventy diseases simultaneously, a scale person-level contact-tracing data does not reach. Section 7 returns to this distinction in assessing what the results below can and cannot support.

1.2 Objectives

The primary objective of this work is to construct and analyze a network model of infectious disease spread between countries that accounts for the geographic distance separating them. This makes it possible to identify which countries play disproportionate roles in disease propagation, to track how those roles shift over time, and to test directly whether geographic proximity makes disease co-reporting more likely. The result informs disease surveillance and intervention strategy: knowing which countries function as structural bridges, and whether that function depends on geography, has direct bearing on where surveillance resources are likely to matter most.

2 Related Work

Network science has previously been applied to infectious disease outbreak patterns. Torres Munguía et al. [8], whose dataset underlies this analysis, characterized the spatial distribution of outbreak-prone diseases and identified clusters of elevated susceptibility, establishing that geographic cluster-

ing exists in this dataset without testing whether that clustering is attributable to distance itself or to a correlated factor. Kim et al. [9] used Monte Carlo simulation to assess epidemic spread risk and concluded that reducing human contact is an effective lever for curbing transmission, working from a contact-network framing rather than a country-level reporting network. Neither study combines temporal centrality evolution with geospatial structure in the same network, and neither subjects a distance-based hypothesis to a permutation test capable of accounting for the non-independence of country-pair observations.

This paper advances that line of work in three respects. First, it tracks centrality and k-core structure across the full 1996–2022 period rather than at a single cross-section, revealing how connectivity responds to globally synchronized events such as the COVID-19 pandemic. Second, it replaces the qualitative observation that nearby countries report diseases together more often with a Mantel permutation test, the correct tool given that country-pair distances and co-reporting counts are not independent draws. Third, it tests whether that distance effect is itself attributable to a known confound, air-travel connectivity, using a partial Mantel test, and stratifies the result by disease transmission mode, a level of statistical scrutiny not present in the descriptive network studies that precede it.

3 Problem Definition

Identifying which countries occupy structurally central positions in a disease-reporting network, and whether that structure is predictable from geography, has direct consequences for surveillance design. A public health body deciding where to place limited surveillance capacity, or which countries’ reporting delays would propagate furthest through the international system, needs to know whether central position is stable over time or shifts with events such as a pandemic, and whether proximity to other high-reporting countries is enough to predict a country’s structural importance or whether that relationship only holds once shared travel infrastructure is accounted for. Neither question can be answered by the descriptive clustering result already established in the literature reviewed above; both require treating centrality and distance as quantities to be tracked over time and tested statistically, which motivates the two questions this study addresses.

First, which countries played key structural roles in disease spread, and how has that role changed over time? Second, does geographic distance shape the pattern of disease co-reporting between countries, and can that relationship be established statistically rather than observed only qualitatively, and does it survive once air-travel connectivity is controlled for? Two analytical components address these questions in turn. Temporal analysis constructs a time-sliced network for each year in the study period and tracks how centrality measures evolve, revealing shifts in which countries occupy structurally important positions as the network grows and changes. Geospatial analysis constructs a weighted network in which edge weight reflects the physical distance between countries, then tests whether that weighting predicts co-reporting behavior, first descriptively, through clustering and spatial degree, and then statistically, through the permutation tests developed in Section 4.

4 Methodology

4.1 Network Construction

Two countries are connected by an edge if they have both reported the same disease in the same year. An edge therefore captures shared exposure to a reporting event; it does not claim that the disease was transmitted directly between the two countries, since the dataset records outbreak occurrence, not transmission pathways. For temporal analysis, the network is sliced into one graph

per year. For the aggregate analyses in Section 6, edges accumulate weight equal to the number of distinct disease-year combinations the two countries share, so that countries with a long history of overlapping outbreaks are more strongly connected than countries that happened to share a single event.

4.2 Centrality Measures

Four centrality measures characterize node importance, each capturing a different aspect of a country’s position in the network. Degree centrality counts a country’s direct connections, weighted by connection strength in the weighted network, and reflects how broadly a country’s reported diseases overlap with other countries’ reported diseases. Betweenness centrality counts how often a country lies on the shortest path between two other countries; a country with high betweenness functions as a structural bridge, since removing it would lengthen the path between many pairs of other countries, corresponding in the disease context to a potential bottleneck or conduit for transmission. Closeness centrality measures how few steps separate a country from every other country in the network, with high closeness indicating a country positioned to reach, or be reached by, the rest of the network quickly. Eigenvector centrality weights a country’s connections by the importance of the countries it is connected to, so that a link to a highly central country contributes more than a link to a peripheral one, identifying countries that are influential by virtue of who their connections are with, not just how many they have.

This project emphasizes degree and betweenness centrality in the year-by-year analysis of Section 6.1, since they map most directly onto the questions of connectivity and critical transmission pathways that motivate this study; closeness and eigenvector centrality are reported for completeness and discussed where they diverge informatively from the first two.

4.3 K-core Decomposition

K-core decomposition simplifies a network by iteratively removing nodes with fewer than k connections, peeling away the periphery to reveal the most densely interconnected core. The resulting core number for each node indicates how deeply embedded that node is in the network’s most connected substructure, and the maximum core number across the network summarizes how resilient the network’s core is to the loss of individual nodes [10].

4.4 Statistical Validation Framework

Section 6 produces descriptive impressions: that connectivity surges during pandemics, that the network has a deeply interconnected core, that nearby countries seem more strongly linked. The third of these, the relationship between geographic distance and co-reporting strength, is directly testable using a Mantel test [11].

The Mantel test is the appropriate tool because the two quantities being compared, the matrix of pairwise geographic distances and the matrix of pairwise co-reporting strengths, are both defined over the same 233 countries. A standard correlation test assumes independent observations, but every entry in a distance matrix shares a row or column with every other entry involving the same country, so the $\binom{233}{2}$ country pairs are not independent draws. The Mantel test builds an empirical null by repeatedly permuting the row and column labels of one matrix, which preserves each matrix’s internal structure while breaking any underlying association between the two, and asks how extreme the observed correlation is relative to that empirical null. We compute the Pearson correlation between the upper-triangular entries of the two matrices, generate a null distribution

from 9,999 label permutations, and report the resulting two-sided permutation p -value. A Spearman version is also reported as a check against a relationship that is monotonic but not linear.

A significant Mantel correlation does not rule out that distance and co-reporting are both driven by a third factor. Countries that are geographically close are also often well connected by direct flights, and flight connectivity is plausibly tied to shared trade, tourism, and surveillance-reporting practices that could independently produce correlated reporting behavior. We extend the test to a partial Mantel test [12], which asks whether distance and co-reporting remain correlated once their shared association with a third matrix, here a country-pair air-travel connectivity matrix built from flight route data, is removed. The partial correlation is

$$r_{XY \cdot Z} = \frac{r_{XY} - r_{XZ} r_{YZ}}{\sqrt{(1 - r_{XZ}^2)(1 - r_{YZ}^2)}} \quad (1)$$

where X , Y , and Z are the vectorized upper-triangular entries of the distance, co-reporting, and connectivity matrices, respectively. Significance is assessed by permuting the row and column labels of the distance matrix X only, holding Y and Z fixed, and recomputing the partial correlation at each of 1,999 permutations, the scheme recommended for partial Mantel tests because permuting X alone tests whether the X - Y relationship persists beyond what X shares with Z , without disturbing the underlying Y - Z relationship being controlled for [13]. We additionally repeat this partial Mantel procedure within three disease transmission-mode strata, vector-borne, respiratory, and waterborne/foodborne (Section 5.3), to test whether the distance effect net of travel connectivity differs by how a disease spreads.

5 Data

5.1 Dataset Description

Two datasets support this analysis. The first, sourced from Torres Munguía et al. [8], records seventy distinct infectious diseases reported across 233 countries from January 1996 to March 2022, where a unique outbreak is defined as at least one reported case of a specific disease in a given country within a given year. The second provides latitude and longitude for 245 countries [14]; pairwise distances are computed with the Haversine formula, which gives the great-circle distance between two points on the Earth’s surface from their coordinates, assuming an Earth radius of 6,371 kilometers [15]. This distance convention is consistent with standard practice in the gravity-model literature on geographic distance, including the GeoDist database [16].

5.2 Exploratory Data Analysis and Network Construction

Exploratory analysis identified and removed six duplicate entries for one country spanning 2020 to 2022, leaving 2,221 entries. Table 1 summarizes the ten most frequently reported diseases over the study period; COVID-19 and influenza dominate by a wide margin, each reported far more often than the next most common disease, cholera. Fig. 1 maps the total number of unique diseases reported by each country across the full study period, showing the United States, China, and several Western European countries reporting the broadest diversity of diseases, a pattern that likely reflects surveillance capacity as much as underlying disease burden.

Table 1: Ten Most Frequently Reported Diseases, 1996–2022

Disease	Count
COVID-19	659
Influenza	409
Cholera	112
Acute Poliomyelitis	112
Yellow Fever	104
Meningitis	95
Measles	83
MERS	67
Dengue	50
Viral Haemorrhagic Fever	49

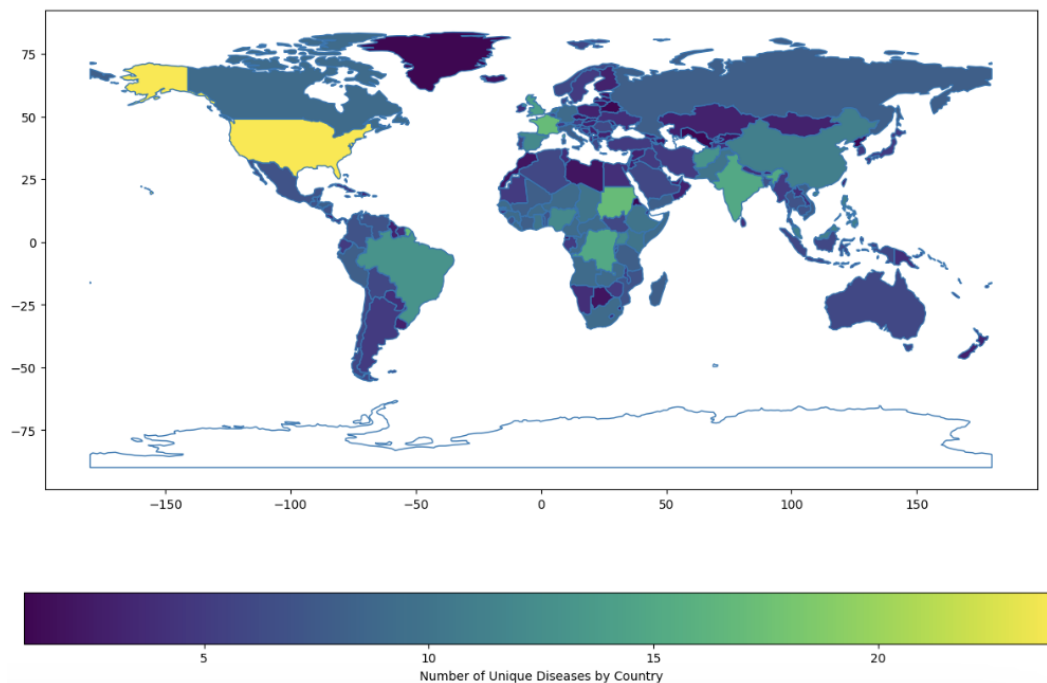


Figure 1: Total number of unique diseases reported by each country, 1996–2022.

The resulting aggregate network comprises 233 nodes and 26,203 edges, with the disease and year of each shared report stored as edge attributes. The top countries by degree and closeness centrality are the United States, China, France, and Germany, with many other countries tied at the same values; the top countries by betweenness centrality are Germany, Russia, Sudan, the Democratic Republic of the Congo, and the Republic of the Congo. These two rankings already hint at the distinction developed further below: degree and closeness identify countries with broad diagnostic and reporting capacity, while betweenness identifies countries whose position in the network makes them structural bridges regardless of their own report volume.

5.3 Connectivity and Disease Transmission-Mode Data

The partial Mantel test in Section 4 requires a third matrix: a measure of direct air-travel connectivity between each pair of countries, independent of co-reporting. We build this from OpenFlights’ public airport and route data [17], mapping each route’s source and destination airports to a country and counting the number of distinct scheduled routes connecting each pair of countries, summed over both directions. This is a coarse proxy: it counts route legs, not passenger volume, and 1,078 of 67,663 routes (1.6 percent) could not be resolved to a country because of naming mismatches between the airport records and the outbreak dataset’s country codes and were dropped rather than guessed. Of the $\binom{233}{2} = 27,028$ country pairs in the network, 2,271 have at least one direct route, a sparsity consistent with the hub-and-spoke structure of real global aviation networks.

Disease transmission mode is assigned per disease from CDC and WHO disease-page descriptions of how each pathogen is primarily transmitted, sorting the seventy diseases in the dataset into vector-borne (nineteen diseases, including malaria, dengue, and yellow fever), respiratory (eleven diseases, including influenza, COVID-19, and measles), and waterborne/foodborne (thirteen diseases, including cholera, typhoid, and acute poliomyelitis). Fourteen diseases with a known but different transmission route (direct contact, bloodborne, or zoonotic-environmental, including Ebola, HIV, and rabies) are retained as a labeled but separate category and excluded from the three-way stratified test since they do not share a common transmission mechanism with each other. Thirteen diseases recorded with an unspecified causative agent, or that are not conventionally transmissible person-to-person at all (for example Creutzfeldt-Jakob disease), are left unclassified rather than assigned a route the data cannot support.

6 Results and Discussion

6.1 Temporal Analysis

Constructing a separate network for each year and tracking average centrality over time reveals two sharp departures from an otherwise noisy baseline: a peak around 2010 and a sustained surge from 2020 through 2022.

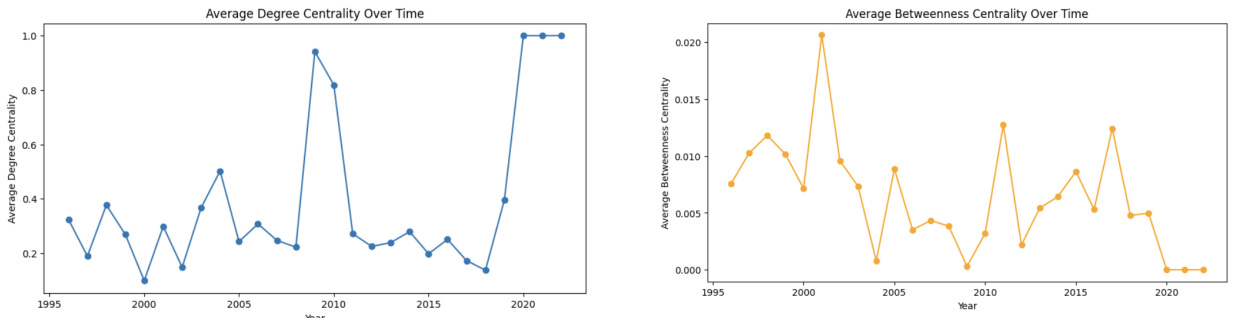


Figure 2: Average degree centrality (left) and average betweenness centrality (right) over time.

Fig. 2 (left) shows average degree centrality rising and falling between 1996 and 2019 without a clear trend, then jumping sharply in 2020. This jump corresponds directly to COVID-19, which was reported by nearly every country in the network within the same year, mechanically inflating connectivity because the network construction in Section 4 links any two countries that share a reported disease in a given year.

Average betweenness centrality (Fig. 2, right) tells a different story. Despite the connectivity surge of 2020–2022, betweenness centrality falls to nearly zero in those years. When almost every country is connected to almost every other country through the same globally reported disease, no single country acts as a bridge, since most pairs of countries are already directly connected. The peak in 2001, by contrast, indicates a year in which disease spread more through indirect chains, where certain countries mediated the connection between others. This contrast between degree and betweenness during the pandemic years illustrates a general property of network structure: a network can become highly connected without becoming more dependent on any particular node, and the two centrality measures together distinguish those situations in a way that degree alone cannot.

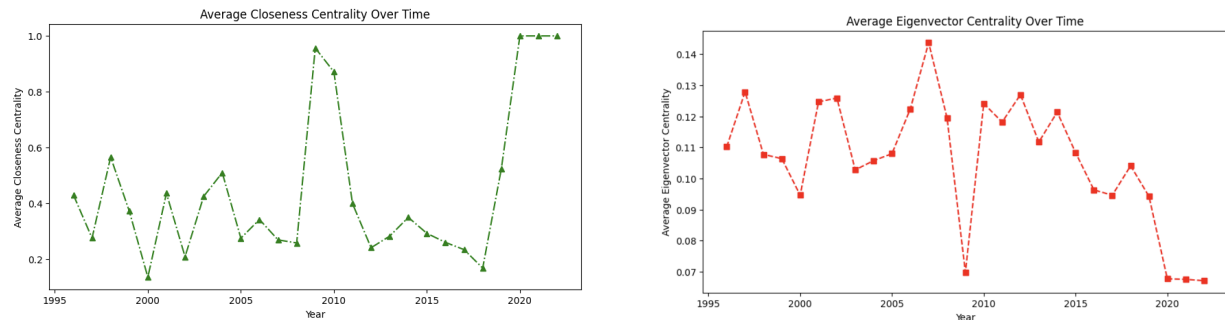


Figure 3: Average closeness centrality (left) and average eigenvector centrality (right) over time.

Closeness centrality (Fig. 3, left) follows degree centrality’s pattern, rising sharply during 2009–2010 and again in 2020–2022, consistent with a network in which most countries are within one or two steps of each other during high-connectivity years. Eigenvector centrality (Fig. 3, right) moves in the opposite direction during the pandemic years, falling to its lowest values in the study period. Eigenvector centrality rewards connections to already-important countries, and when connectivity becomes nearly uniform across the entire network, the distinction between connecting to an important country and connecting to any country disappears, compressing the eigenvector centrality distribution toward uniformity.

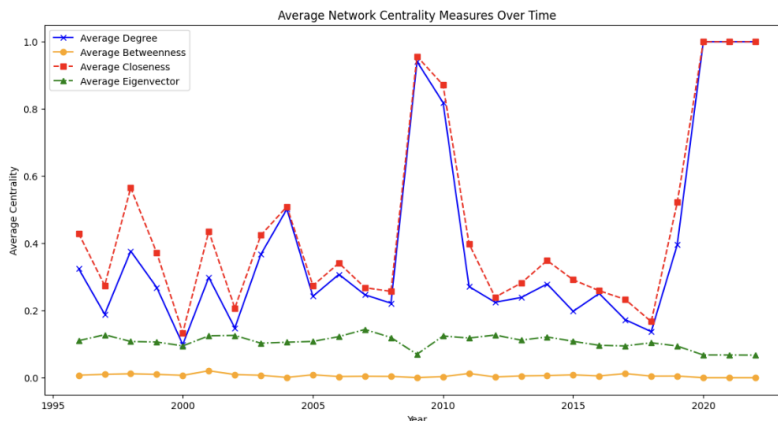


Figure 4: All four average centrality measures over time, shown together.

Fig. 4 places all four measures on the same axes. Degree and closeness move together throughout the study period, while betweenness and eigenvector centrality stay low and comparatively flat, with eigenvector centrality declining further still during the pandemic years. The overall picture is a network whose connectivity is widespread rather than concentrated in a small set of hub countries, a structural property the k-core analysis below examines more directly.

Tracking degree centrality for a fixed set of countries across different regions, the United States, Brazil, Australia, Colombia, China, Sudan, and Russia, shows substantial heterogeneity that the global averages above obscure.

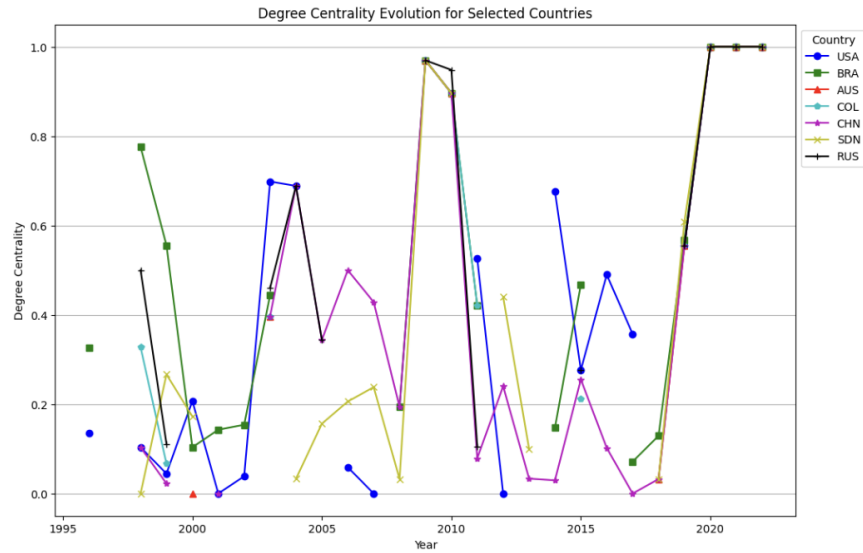


Figure 5: Degree centrality evolution for selected countries, with years of no reported disease left as gaps.

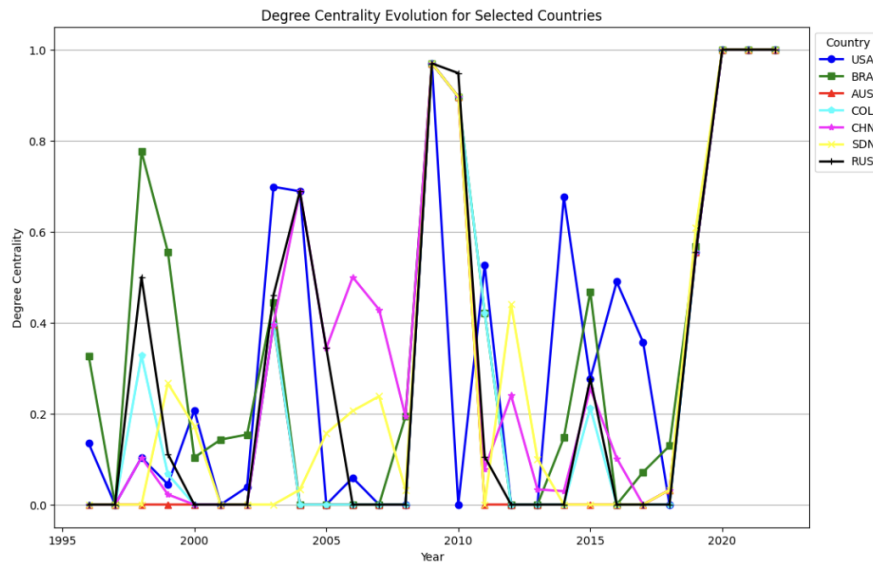


Figure 6: Degree centrality evolution for selected countries, with years of no reported disease set to zero.

These two treatments of missing reports answer different questions. The gap version (Fig. 5) shows each country’s centrality conditional on having reported at least one disease that year, while the zero-filled version (Fig. 6) shows centrality unconditionally, including years of apparent inactivity. The difference between the two figures measures how much a country’s centrality trajectory is driven by gaps in reporting rather than by an actual absence of disease activity, a distinction that matters for interpreting any single country’s trajectory.

Because the average centrality measures above peak sharply around 2010, we examine 2008–2010 in more detail by constructing the network for each of those years individually.

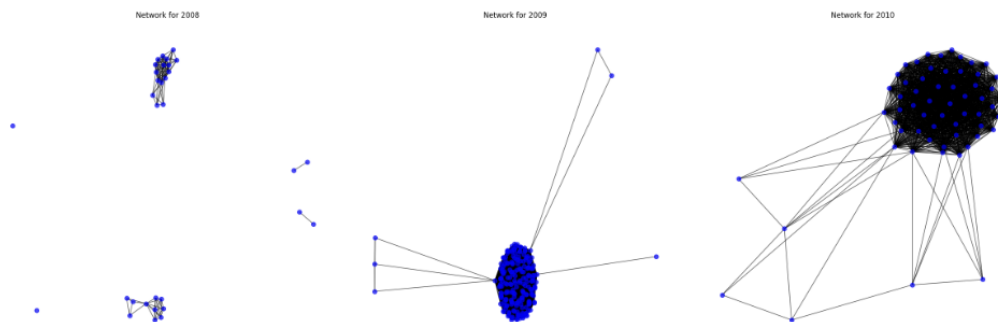


Figure 7: Network structure in 2008, 2009, and 2010, showing the rapid densification of the disease-reporting network over three years.

Fig. 7 shows isolated clusters in 2008 consolidating into a single dense core by 2010. The disease driving this consolidation can be identified by decomposing the 2010 network into subgraphs for each individual disease reported that year.

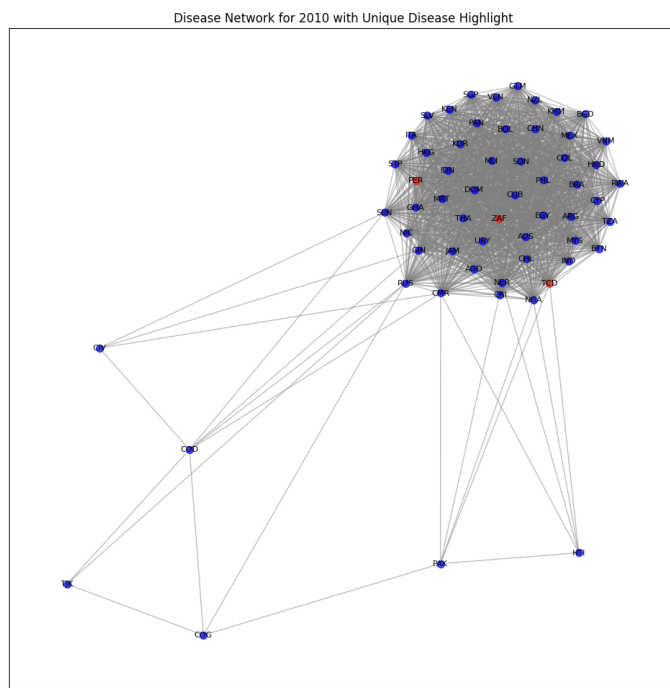


Figure 8: The full 2010 disease-reporting network, with countries that reported an otherwise-isolated disease case highlighted.

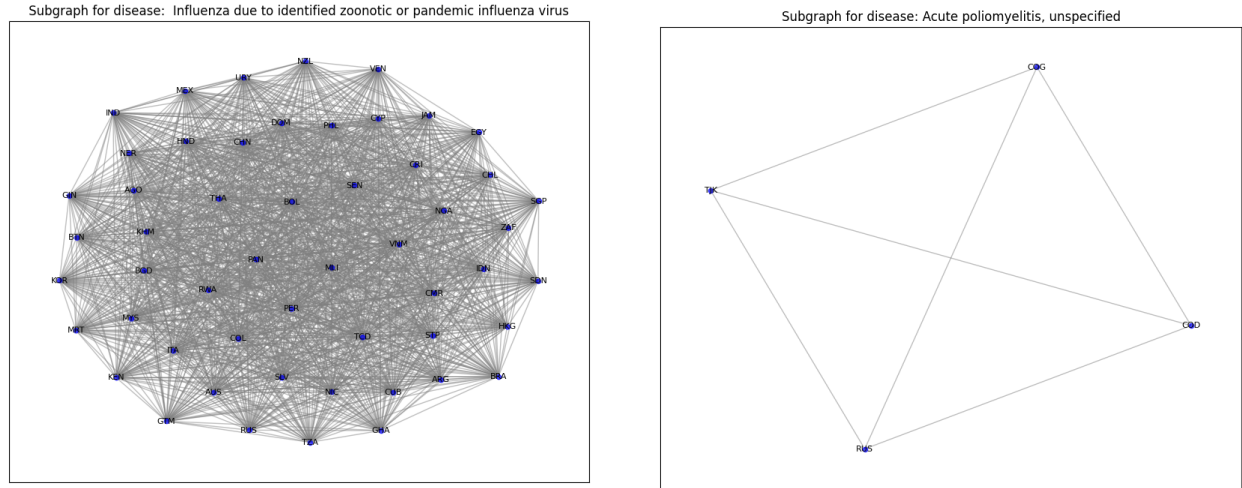


Figure 9: Subgraphs for two diseases reported in 2010: influenza (left), forming a near-complete graph among dozens of countries, and acute poliomyelitis (right), connecting only four. The contrast illustrates how disease-specific reporting patterns range from globally synchronized to highly localized within the same network and the same year.

To assess the network’s resilience to the loss of individual nodes, we aggregate disease reports across all years into a single weighted network, with edge weight equal to the cumulative number of shared disease-year reports between each pair of countries, and apply k-core decomposition.

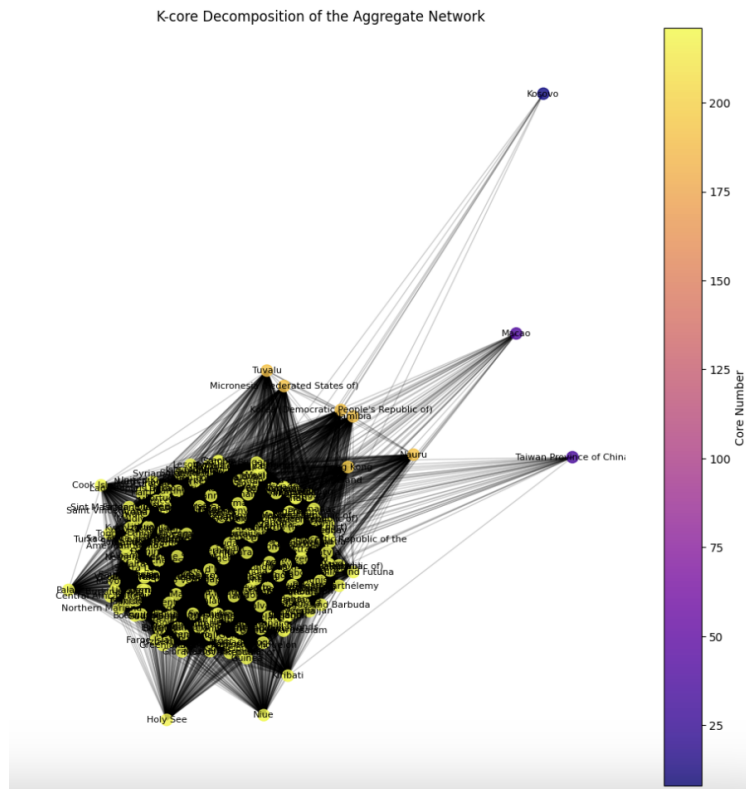


Figure 10: K-core decomposition of the aggregate network. Color indicates each node’s core number.

The maximum core number is 221, and the great majority of nodes share this value, indicating a network whose core is deeply and almost uniformly interconnected rather than concentrated around a small set of hubs, consistent with the pattern already visible above: the network’s high connectivity is broadly distributed, not centrally coordinated. Table 2 reports centrality scores for a sample of nodes at the top and bottom of the distribution. Canada, China, Hong Kong, Iran, and Israel sit at or near the top by degree and closeness centrality, while Taiwan, Macau, and Kosovo sit near the bottom, with Kosovo’s degree centrality more than an order of magnitude lower than the network median. These low values reflect each territory’s limited diplomatic and reporting infrastructure within the WHO surveillance system rather than an absence of disease activity, an important caveat for interpreting any centrality ranking built from international reporting data.

Table 2: Centrality Scores for Selected High- and Low-Centrality Nodes

Country	Degree	Betweenness	Closeness
Canada (CAN)	0.9914	2.38e-04	0.9915
China (CHN)	0.9957	4.14e-04	0.9957
Hong Kong (HKG)	0.8836	3.31e-04	0.8958
Iran (IRN)	0.9871	4.30e-05	0.9872
Israel (ISR)	0.9914	2.16e-04	0.9915
Palau (PLW)	0.9569	1.69e-07	0.9587
Niue (NIU)	0.9526	0.0000	0.9547
Taiwan (TWN)	0.1595	0.0000	0.5433
Macau (MAC)	0.1767	0.0000	0.5485
Kosovo (XXK)	0.0345	0.0000	0.5088

6.2 Geospatial Analysis

This section integrates geographic distance directly into the network by weighting each edge according to the distance between the two countries it connects, with weight defined as the inverse of distance so that closer countries receive higher weight.

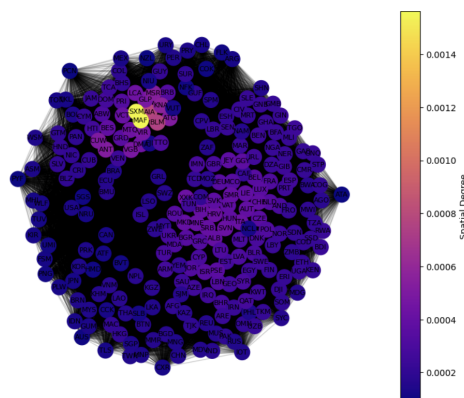


Figure 11: Network visualization colored by spatial degree, the average connection strength of each node to its geographic neighbors.

Table 3 lists the five countries with the highest average connection strength to their geographic neighbors. All five are small Caribbean island territories, Saint Martin, Sint Maarten, Anguilla,

Saint Barthélemy, and Saint Kitts and Nevis, whose high spatial degree reflects geographic proximity to a cluster of similarly small neighboring territories rather than any particular disease burden.

Table 3: Top Five Countries by Geographic Connection Strength

Country	Strength
Saint Martin	0.389
Sint Maarten	0.386
Anguilla	0.193
Saint Barthélemy	0.187
Saint Kitts and Nevis	0.142

The weighted clustering coefficient measures how interconnected a node's neighbors are with each other, identifying tightly-knit regional clusters.

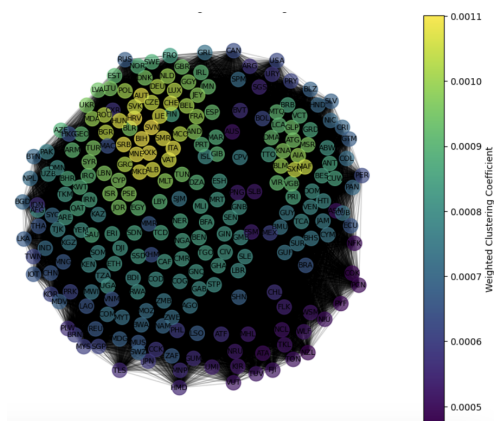


Figure 12: Network visualization colored by weighted clustering coefficient.

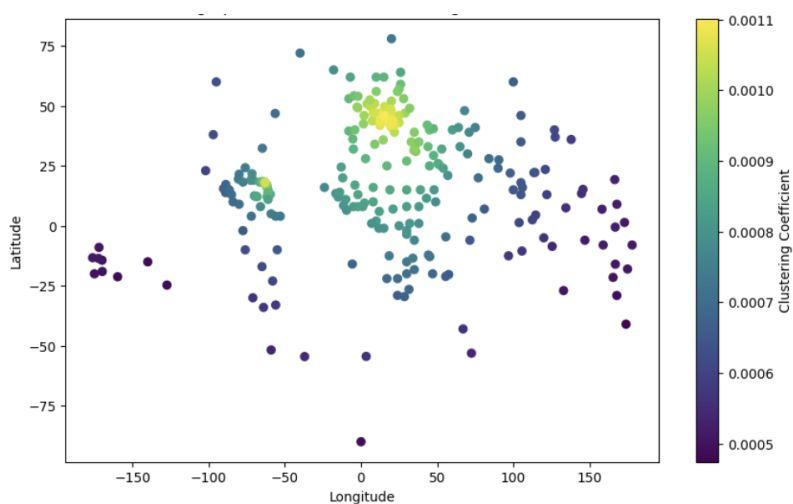


Figure 13: Weighted clustering coefficients mapped onto geographic coordinates.

Mapping the clustering coefficient onto each country’s actual location (Fig. 13) shows the highest values concentrated in Western and Central Europe and in the Caribbean, both regions of small countries packed closely together, and the lowest values among isolated Pacific island nations. This geographic concentration of high clustering coefficients is the descriptive observation tested statistically below.

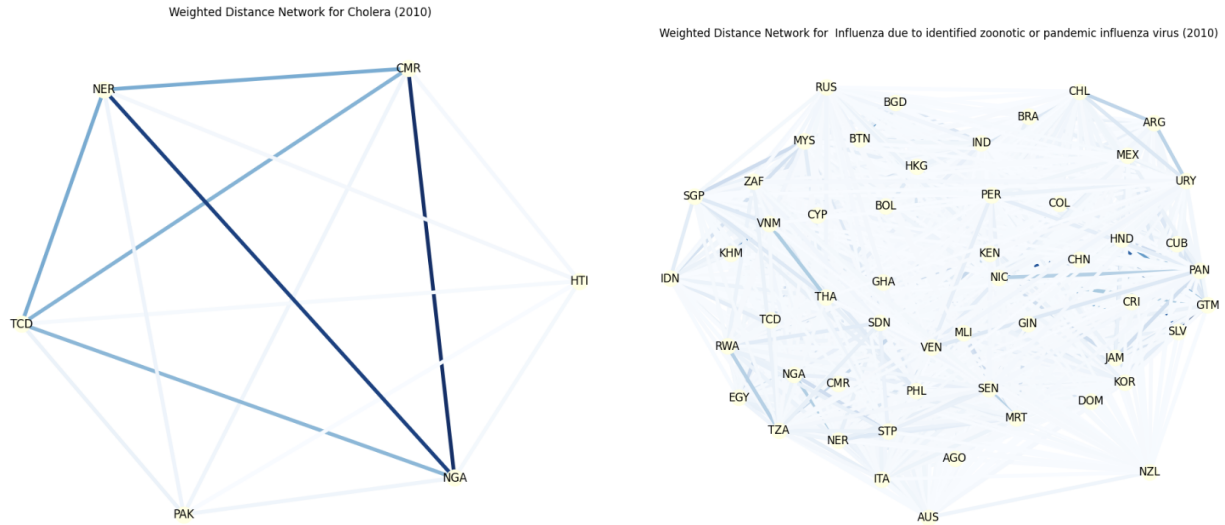


Figure 14: Weighted distance subgraphs for cholera (left) and influenza (right) in 2010. Darker edges indicate shorter distances. Cholera’s 2010 reports connect a small, geographically dispersed set of countries with no consistent proximity pattern, while influenza’s far larger and more globally distributed 2010 reports show the weighted network at a scale where any individual proximity relationship is visually difficult to assess, motivating the network-wide statistical test below rather than visual inspection of individual disease subgraphs.

6.3 Statistical Validation of the Distance Effect

The clustering map in Fig. 13 and the connection-strength results above are consistent with geographic proximity shaping disease co-reporting, but consistency is not confirmation. Applying the Mantel test from Section 4 to the full 1996–2022 network, restricted to the 233 countries with both outbreak and coordinate data, gives a Pearson correlation of $r = -0.192$ between geographic distance and co-reporting strength, with a permutation p -value below 0.0001 across 9,999 permutations (Fig. 15). The Spearman correlation, included as a check against a relationship that is monotonic but not linear, is $\rho = -0.145$, also significant at $p < 0.0001$. The negative sign confirms the direction the descriptive analysis suggested: countries that are geographically closer share more disease-year reports.

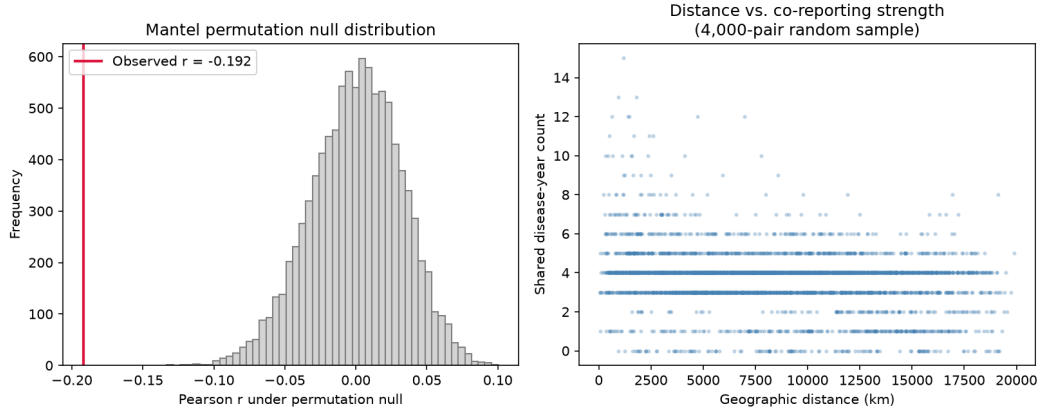


Figure 15: Mantel permutation test results. Left: the observed correlation (red line) against the null distribution generated by 9,999 label permutations. Right: a random sample of 4,000 country pairs showing the underlying relationship between distance and co-reporting count.

The effect size is modest: distance accounts for roughly four percent of the variance in co-reporting strength by Pearson’s r^2 . A single globally reported disease such as COVID-19 connects nearly every country regardless of distance, which dilutes any distance-dependent signal coming from more geographically localized diseases. A dilution account of this kind predicts that excluding the pandemic years should strengthen the correlation. The data show the reverse: restricting the analysis to years before 2020 weakens the correlation ($r = -0.107$, $p < 0.0001$). The timing of a country’s COVID-19 report, even though nearly every country eventually reported the disease, still followed the geographic diffusion of the pandemic itself, so that neighboring countries reported in closer temporal proximity to each other than distant countries did, strengthening the distance signal during 2020–2022 rather than washing it out. This result depends on comparing the statistical association across two overlapping but distinct time windows, and motivates disaggregating the network by disease, taken up next, to clarify which diseases drive the distance effect.

The statistical significance is unambiguous despite the modest effect size, converting the qualitative impression from Fig. 13 into a tested claim: geographic distance has a small but statistically robust association with which countries co-report infectious diseases.

6.4 Robustness to Travel Connectivity and Disease Transmission Mode

The Mantel test above establishes that distance and co-reporting are correlated, but not that distance does this explanatory work itself rather than standing in for travel connectivity between well-connected countries. Applying the partial Mantel procedure from Section 4, controlling for the air-travel connectivity matrix described in Section 5.3, the correlation between distance and co-reporting weakens only slightly, from $r = -0.192$ unconditionally to a partial $r = -0.176$ ($p < 0.001$ across 1,999 permutations). Distance and connectivity are themselves only weakly correlated ($r = -0.105$): geographically close countries are somewhat, but not strongly, more likely to have a direct flight route between them. Distance survives as a predictor of co-reporting in its own right, not as a proxy for which countries happen to fly to each other.

Table 4: Partial Mantel Test Results, Controlling for Air-Travel Connectivity

Stratum	Countries	Pairs	Partial r	p
Full network	233	26,203	-0.176	< 0.001
Vector-borne	108	735	-0.070	< 0.001
Respiratory	232	26,180	-0.144	< 0.001
Waterborne/food.	105	1,402	-0.111	< 0.001

Stratifying this confound-controlled test by transmission mode (Table 4) produces a result that runs against the intuition that vector-borne diseases, being tied to the geographic range of a mosquito, tick, or other arthropod vector, should show the strongest distance effect of the three. Instead, vector-borne diseases show the weakest distance-controlled correlation ($r = -0.070$), while respiratory diseases show the strongest ($r = -0.144$), with waterborne/foodborne diseases in between ($r = -0.111$).

This pattern is consistent with a distinction established in spatial ecology between two sources of distance decay: decay driven by dispersal limitation, where similarity falls off with distance because organisms, or in this case pathogens, cannot easily reach far-away locations, and decay driven by environmental filtering, where similarity instead tracks an environmental gradient that need not itself be spatially compact [18]. Most vector-borne diseases in this dataset, including malaria, dengue, and yellow fever, are climatically constrained to the global tropical and subtropical belt, a habitat band that circles the planet rather than occupying a single contiguous region; Brazil, Nigeria, and Indonesia are all climatically suitable for these vectors while being geographically remote from one another. Climate is a primary determinant of vector-borne disease distribution, and that distribution is increasingly shaped by climate-driven range expansion independent of simple geographic proximity [7]. If vector-borne co-reporting is governed primarily by shared climate suitability rather than shared location, environmental filtering by climate zone dilutes the distance-decay signal relative to diseases whose spread depends more directly on regional contact and mobility patterns, such as respiratory and waterborne/foodborne disease, which is the pattern observed here. Testing this mechanism directly would require climate-zone data this analysis does not include, a natural target for the future work discussed in Section 7.

7 Limitations and Future Work

The Mantel and partial Mantel tests above test the role of geographic distance directly rather than asserting it, and the result holds after controlling for travel connectivity. The structural limitation introduced in Section 4, that this network connects countries rather than people, is not addressed by further statistical testing. "Spread" at the country level means shared exposure to a reporting event, not a claim about how a pathogen moved between two individuals. Additional confound control narrows which alternative explanations remain plausible; it does not change what an edge in this network represents. The results above should be read as a country-level surveillance and pattern-detection signal, not a transmission-chain reconstruction.

Within that scope, narrower limitations remain. The dataset lacks node-level epidemiological detail, case counts, severity, or case fatality, that would let centrality and clustering results be weighted by health impact rather than by report count alone. Several disease-specific subgraphs in Fig. 9 and similar disease-year combinations not shown form complete graphs among a small number of countries, which limits how much can be inferred about transmission structure from those subgraphs individually. The air-travel connectivity matrix counts route legs rather than passenger

volume and leaves 1.6 percent of routes unresolved to a country, and the disease transmission-mode classification leaves thirteen diseases unclassified and a further fourteen in a labeled but heterogeneous "other known route" category excluded from the stratified test.

Future work has two directions. The first stays within the current network: testing additional candidate confounds, such as trade volume or shared official language, against the partial Mantel framework already built here, and directly testing the climate-zone explanation proposed above for the weak vector-borne distance effect, which this paper proposes but does not verify. The second direction changes scope rather than extending the current network: moving from country-level co-reporting to mobility-weighted or genomic/phylogenetic transmission data would address the structural limitation above directly, at the cost of either narrowing to a single well-sequenced pathogen or requiring mobility data not available to this project.

8 Conclusion

This paper analyzed the global network of infectious disease co-reporting along temporal and geospatial dimensions. The temporal analysis showed that network connectivity is highly sensitive to globally synchronized events, rising sharply during the COVID-19 pandemic, while remaining structurally distributed rather than concentrated around a small set of hub countries, evidenced by a maximum k-core of 221 shared by most of the network. The geospatial analysis showed that geographic proximity is associated with stronger disease co-reporting connections, a relationship the Mantel permutation test confirms is statistically significant and not an artifact of the network's dependency structure, that survives controlling for air-travel connectivity in a partial Mantel test, and that holds, and strengthens, when the pandemic years are included. Stratifying the confound-controlled test by transmission mode showed the distance effect is weakest for vector-borne disease and strongest for respiratory disease, a pattern consistent with environmental filtering across a global climate zone overriding simple geographic proximity, a mechanism this paper proposes from established ecological theory but does not test directly within this dataset. These results move the central claim motivating this study, that distance matters for disease co-reporting patterns, from a visual observation to a tested and confound-checked finding. The network constructed here remains a country-level reporting signal rather than a person-to-person contact network, and testing additional confounds and verifying the climate-zone mechanism directly are the most promising directions for future work.

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Code and Data Availability

All code, data, and the statistical validation analysis supporting this paper are openly archived with a permanent DOI at Zenodo: Oje, S.A. (2026). Analyzing the Network of Disease Distribution by Country and Distance. Zenodo. [https://doi.org/10.5281/zenodo.\[XXXXX\]](https://doi.org/10.5281/zenodo.[XXXXX]). This includes the network construction and visualization notebook, the plain and partial Mantel test scripts, the

air-travel connectivity construction script, and the disease transmission-mode classification. Code is released under the MIT License; the paper and figures are released under CC BY 4.0.

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